The Frozen Frame

An Instrument That Measures a Language Model
Against a Ruler Made From Itself

Built by Claude instances, steered by a human, measured by machines
Toolkit: Heinrich

The first Heinrich paper whose figures were captured, not drawn—
each rendered from the same run, in the same frame, that produced its numbers.

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Abstract

The previous Heinrich papers were about what models do. This one is about how to know. They shared a hidden dependency: every geometric claim was measured against a coordinate system that the measurement could itself move. A safety direction, a displacement, a trajectory—each was read off a frame re-derived from the data at hand, so the ruler flexed with the thing it measured. This paper describes the instrument built to stop that. A per-layer PCA frame is computed once over the model's own vocabulary and never re-derived; the number of retained components equals the hidden width and the components are orthonormal, so distance in the score coordinates *is* distance in the hidden state, exactly. Every token—any of the tens of thousands—then holds a fixed address, and any live forward pass can be measured against all of them at once with no truncation anywhere in the metric. The instrument declares where it cannot be trusted: a per-layer noise floor from `float16` forward reproducibility, and a frame-validity test by principal angles against a held-out projection. And it renders every measurement as a navigable surface an agent can drive and photograph, so a figure in this paper and the number beside it come from one session and one frame. We demonstrate the instrument on a single question—during a forward pass, does the residual migrate toward the token it will emit?—and answer it: no. The state does not travel to the answer; it rotates to aim at it. We close with what the instrument cannot do, and why that ceiling is a property of the method, not of the hardware.

Contents

1 What This Document Is

The prior papers in this series measured behaviour: which models refuse, which break, how safety is arranged in the residual stream. HEINRICH v3 ended on a confession—every measurement error it had made pushed in the same direction, making models look safer than they were, because the investigator shared training lineage with the investigated. The lesson was not "we found the right number this time." The lesson was that a measurement instrument must be able to state its own bias, or it will quietly encode the measurer's.

This document is the instrument built from that lesson. It is not about safety, or any one model, or any one finding. It is about a ruler: how to hold a language model's internal geometry still enough to measure, how to prove the ruler has not moved, and how to publish a measurement so that its picture and its number cannot drift apart. The finding we use to exercise it—that a model *aims* at its next token rather than *travelling* to it—is a worked example, chosen because it is the cleanest thing the ruler has shown us, not because the paper is about it.

A note on the figures. This is the first Heinrich paper with any. Every earlier one was prose and tables because a hand-made figure is a claim you cannot check: the picture is drawn to illustrate a number computed elsewhere, and nothing binds them. Here an agent runs the measurement and then drives the live instrument to photograph the exact state that measurement produced. The picture and the number are the same event. Each figure carries, in monospace beneath its caption, the `.mri` it came from and the script that captured it. None of them are drawn.

2 The Problem: A Ruler That Moves

To compare two hidden states—this layer to that layer, this token to the answer token—you need a coordinate system they share. The usual move is to fit one from the data: PCA the activations you have, read the coordinates off the top components. The trouble is that the frame then depends on *which* activations you fed it. Add prompts, change the sample, re-run the probe, and the axes rotate; a "direction" that was there at n=8 is gone at n=15 (v3 lost its safety layers exactly this way). The ruler is a function of the thing being measured, so every distance is contaminated by the choice of measuring set. Worse for trajectories: to ask whether a live state is *near* a particular token, that token must have coordinates in the same frame—but a token you did not sample has no address at all.

What is required is a frame that (i) is fixed independently of any particular query, (ii) gives every token an address, and (iii) preserves true distances, so that "close in the picture" means "close in the model." The next section is that frame.

The field's standard readout confronts a cousin of this problem from the other side. The logit lens reads a hidden state through the final unembedding and returns gibberish at intermediate layers, because each layer "speaks its own language"; the tuned lens (Belrose et al., 2023) repairs this with a *learned* affine translator fit per layer. That is a second moving part—a trained transform between the state and its reading. The frozen frame takes the opposite route: an exact, trainless change of basis (below) that leaves every distance untouched, so there is nothing to fit and nothing to fit *to*.

3 The Frozen Frame

Capture the model's residual state once, at every layer, over a broad token sample, and decompose each layer with PCA—but retain *all* D components, where D is the hidden width, and store the components C_ℓ, the mean μ_ℓ, and the silence baseline b_ℓ. This frame is written to disk and never recomputed. A live hidden state h_ℓ receives coordinates by the same fixed transform,

$$s \;=\; (h_\ell - b_\ell - \mu_\ell)\, C_\ell^\top, \qquad C_\ell C_\ell^\top = I, \qquad \|s - s'\|_2 \;=\; \|h_\ell - h'_\ell\|_2.$$

Because C_ℓ is a full-rank orthonormal basis, the projection is a lossless rotation: the score-space L_2 distance is the hidden-space L_2 distance, with no truncation approximation anywhere. The frame is not a summary of the state; it is a change of basis for it.

The vocabulary is projected through this same frozen frame—every token, not a sample—so each token id acquires a fixed per-layer address (`vocab_scores`). A live trajectory can therefore be compared, exactly, against the entire vocabulary at every depth in one pass. The frame is the coordinate contract; the `.mri` file is the artifact that carries it between the producer that captured it and any consumer that reads it.

Two subtractions are load-bearing and easy to forget. The stored mean μ_ℓ recentres the frame; the silence baseline b_ℓ removes the model's resting state. Omitting either leaves a constant offset that, at some layers, dwarfs the signal by more than an order of magnitude. Both are applied on every projection.

4 The Instrument Declares Its Limits

A distance is only meaningful above the noise of the machine that produced it. HEINRICH measures that noise directly: it re-captures the sample forward pass in `float16` and compares to the stored scores, giving a per-layer reproducibility floor. Distances at or below the floor are the arithmetic of the hardware, not the model, and are drawn as such.

The frame's own validity is testable, not assumed. A second projection is derived from a held-out, crystal-suppressed vocabulary sample, and its principal angles against the frame are measured layer by layer; the variance-weighted overlap is reported as a falsification statistic. On SmolLM2-135M (raw) the frame holds at 0.90–0.999 at every layer—stated as a licence to trust the ruler, revocable the moment a model fails it.

4.1 Falsified against ground truth

Self-declared trust is cheap unless the instrument can be caught fabricating. On real language models there is no answer key. On a model we *built* there is. We tested the frame against matched causal-bank specimens—11.4M-parameter models, deterministic substrate, pairs differing by exactly one flag (normalisation on/off) at two seeds—built as a null case by the training-side agent. Two tests, both adversarial by construction.

Artifact rate. A re-capture of the same checkpoint is bit-identical (CKA 1.000): the instrument adds no noise of its own. So on a pair that differs by exactly one flag, the entire reported gap is attributable to that flag—nothing beyond the controlled variable is reported. It also orders causes correctly, which a smearing frame would not: the flag effect (CKA $\approx$ 0.59) exceeds the seed-only effect (CKA $\approx$ 0.76), which exceeds noise (0); by displacement correlation the separation is starker still (seed 0.98–0.996, flag 0.69–0.71).

comparison	varies	CKA	disp. corr
re-capture of same checkpoint	nothing (noise floor)	**1.000**	**1.000**
same config, different seed	seed only	0.73–0.79	0.98–0.996
same seed, opposite flag	flag only	0.58–0.60	0.69–0.71

Table 1: The instrument against a ground-truth null. Zero noise floor, and the one controlled variable (the flag) is the dominant reported difference—no structure appears that is not traceable to a known cause.

Does the method paint the cliff onto everything? One tempting check is worthless and we flag it as such: running the depth analysis on a depthless causal bank returns an empty list, but only because the tool finds no per-layer files and stops, having examined no data—it would return empty for any input of the wrong shape. The real control runs the readout-cliff of Section 6 on a transformer, against targets with no reason to commit. For "The capital of France is," the true answer commits at 0.77 depth; of nine implausible words (" bicycle," " velvet," " thermostat," …) not one commits at any layer —each sits near the bottom of the sixteen-word panel from first layer to last. Type-appropriate but wrong tokens (other capital cities) do eventually out-read the random panel, but only at the final layer, never in the band. The three-quarters commit is specific to the answer that fits; the method does not impose it on arbitrary tokens—which is the property an empty list could never have shown.

5 The Shared Surface

The frame is not only a metric; it is a place. HEINRICH's companion renders the frozen frame and any live trajectory as a navigable three-dimensional surface, and it obeys one rule that makes its pictures admissible as evidence: *omit, never fake.* A segment renders only when both endpoints are genuinely in range; a candidate absent from a layer's readout is not interpolated across the gap; every view states its own sampling ("2,000 of 48,660") and its own frame ("rel to live, s=0.52"). The honesty is in the pixels, so a photograph of the surface carries its caveats into the PDF.

This is what lets an agent produce figures that are measurements. The agent drives the same instrument that computed the numbers, frames a view, and captures it in the same session and the same frame. Figure 1 was made exactly this way.

6 Worked Example: Aiming, Not Traveling

We ask the instrument one question. During a forward pass, does the final-token residual move toward the location of the token it is about to emit? The frozen frame makes this answerable exactly: the answer token has a fixed address at every layer, and so does the live state.

The proximity form of the hypothesis is false. Conditioned on the model actually producing the answer, the answer token's own frozen state sits at roughly chance rank among a background panel *even at the final layer*: the trajectory does not go there. The readout form is true. Reading each layer's state through the model's output matrix, the answer out-competes the whole panel from a late band onward and at the final layer without exception. The state does not approach the answer's location; it rotates until it projects onto the answer's output direction. Distance-to-state and preference-to-emit are different geometries—and the frozen frame is what let us hold them apart. Figure 1 is that distinction made visible: a straight spine that never bends toward a token, and a fan of readout candidates collapsing to the commit.

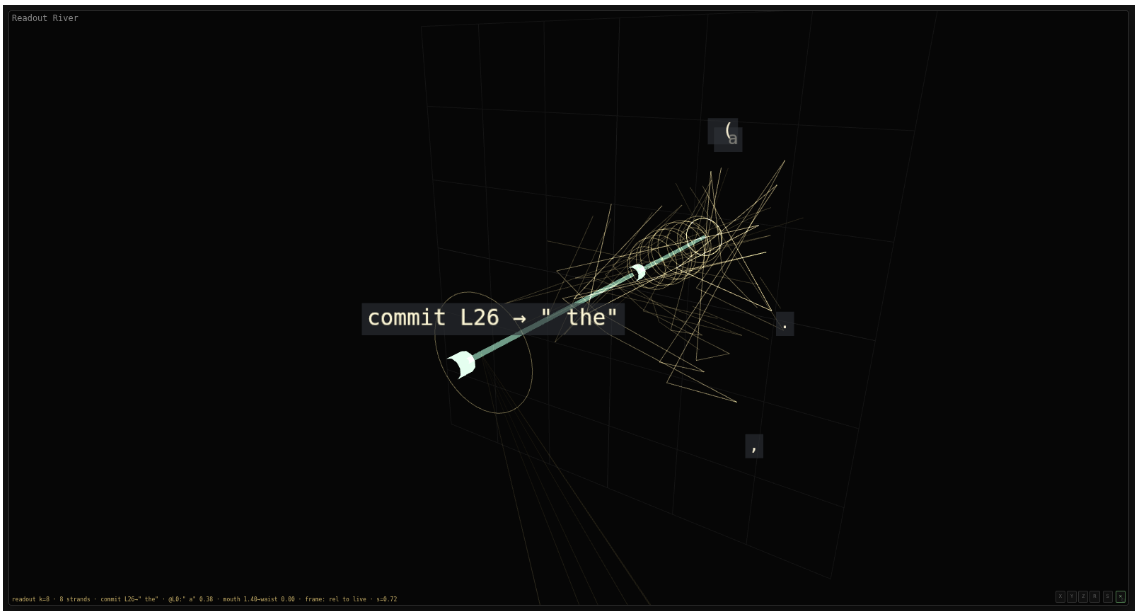

Figure 1: **The readout fan ("the horn").** SmolLM2-135M (base), prompt *"The capital of France is,"* rendered in the frozen frame. The straight teal spine is the live trajectory through depth; it runs down the axis and does not bend toward any token's location. The gold fibers are the per-layer top-k readout candidates, each drawn at its true offset from the live state; they fan wide at the mouth (early layers) and collapse to the commit at the waist (`L26` → " the"). A candidate absent from the top-k between two layers is not bridged—the fiber breaks rather than fake continuity. Labels resolve through the full tokenizer (whitespace shown as a visible glyph, never a raw id).

`mri: web/.data/smollm2-135m/raw.mri    capture: paper/figures/capture_horn.mjs    readout: lmhead @ h/|h|`

One horn is an anecdote. Replaying the identical protocol—twenty-eight completion prompts, a sixteen-token background panel—on three further models, one of them from a different tokenizer and architecture family, turns it into a measurement. The answer out-reads the whole panel at the final layer in twenty-seven to twenty-eight of twenty-eight runs on every model, so the readout commit is not a one-model artifact. And the commit is scheduled by *relative depth*: the answer's rank collapses to zero near three-quarters of the way through the network regardless of layer count—the median commit sits at 0.71–0.75 of depth across models of 24, 30, and 32 layers—and regardless of family; Qwen, a different tokenizer and architecture entirely, commits at the same fraction. Scale buys correctness (the 360M model answers 24 of 28 versus 17 for the base 135M) but not an earlier decision, and instruction tuning barely moves the band. The instrument turned a single picture into a depth-invariant regularity that survives a family boundary (Figure 2).

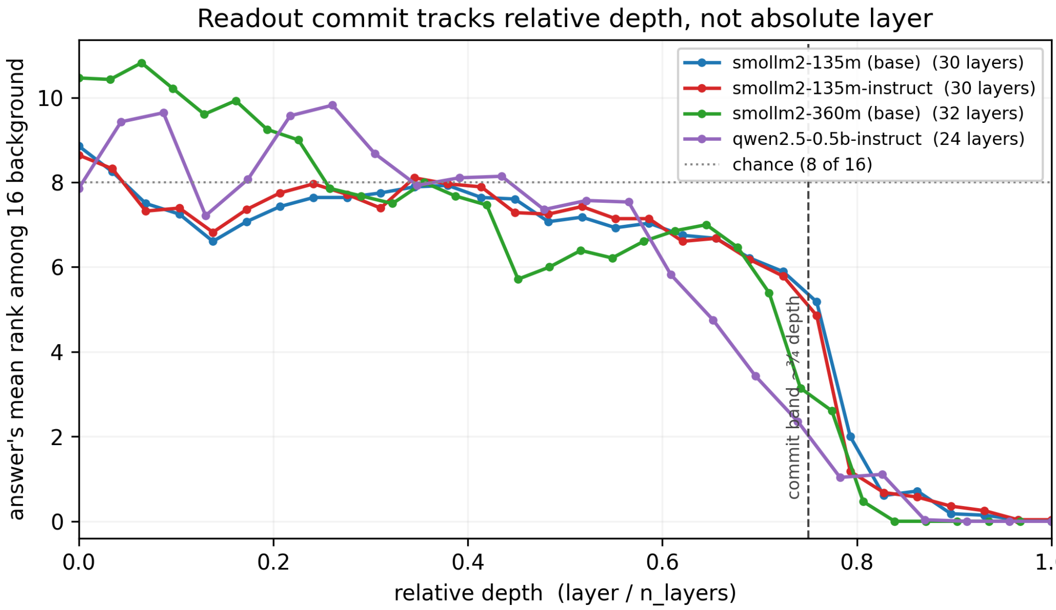

Figure 2: **The readout commit tracks relative depth.** For each model, the answer token's mean rank among the sixteen-token background panel (0 = it out-reads all of them) against relative depth (layer / n_layers). All three sit near chance (8 of 16) through the first two-thirds of the network, then collapse to rank 0 in a narrow band at about three-quarters depth—across 24, 30, and 32 layers and two tokenizer families (SmolLM2 and Qwen), aligned only once depth is normalised. The decision is scheduled by depth fraction, not absolute layer.

`data: docs/data/homing-study-v{3,4}*.json` `run: scripts/homing_study_v4.py` `plot: scripts/figure2_homing_band.py`

This sits inside known structure, and the instrument's role is to sharpen a distinction the standard tools blur. That prediction quality accrues smoothly, at a fixed multiplicative rate per layer, regardless of architecture or scale, is an established law of next-token prediction (arXiv:2408.13442, 2024). Our observation is not that schedule but its complement: a specific answer's *win* over its competitors is not smooth at all but a late phase transition near three-quarters depth, invisible to the aggregate residual that law measures. That the decision is fixed internally well before the output echoes "decide early, explain later" (arXiv:2604.22266, 2026); that the state *selects* rather than approaches its token is the same position-versus-readout question a concurrent geometric account is now asking (Lombardo et al., 2026), answered here with an exact hidden-space distance rather than a lens. The instrument's claim is not a new law; it is that these distinctions become measurable, and falsifiable, once the ruler holds still.

6.1 Where the commit comes from — and the discipline of getting there

Homing says the commit is a readout rotation, not a move; where does the rotation come from? Eight further experiments traced it, and the answer matters less than how the tracing went. The rotation is written by the MLP, not by attention to content. Through the first three-quarters of the network the last token parks most of its attention on a dead position-0 sink — a low-value slot that writes *less* than the content it ignores — and at the commit layer that sink's attention concentrates to 0.9 and becomes the dominant channel that primes an MLP write. The write is diffuse: dozens to ∼150 neurons, no clean answer-specific core even after the generic direction is subtracted, and it does not survive rewording — paraphrase the prompt and different neurons carry it, only a generic capital-slot substrate stable across forms. It generalises (three models commit at 0.74–0.84 depth, all MLP-led), and the larger model writes more cleanly. So a small model does not *retrieve* the fact; it *reconstructs* the answer, per surface-form, from a smeared substrate, on a fixed depth clock, through a register otherwise switched off.

But for a paper about an instrument the point of the chain is Table 2. At every step the first, clean-looking result was an artifact and a control dissolved it; the honest finding always lay one control deeper. The instrument's worth was not the mechanism it found. It was the six answers it refused.

the seductive result	the control that killed it	what survived
an L^* and a clean rank curve	final-layer lens $\neq$ the model's word	a double-norm; use true logits
2.35× attention to the licenser	the licenser *is* the previous token	recency, not meaning
0.79 / 53× at the commit	the licenser *is* the position-0 sink	the sink, not meaning
the MLP writes +77 to the answer	it writes +119 to the null	scale, not specificity
a concentrated top-neuron set	subtract the generic direction	no clean core; diffuse
activity collapses to few neurons	the same neuron for both tokens	a shared crystal, not the token's

Table 2: The instrument catching itself. Six times in one investigation the clean answer was an artifact a control dissolved; the honest finding always lay one control deeper. This is the discipline the frozen frame and the declared floors exist to enforce, made visible.

7 Two Witnesses on a Newborn

The frozen frame and the discipline it enforces are not specific to transformers. The same instrument, aimed at a different body, produced this paper's cleanest demonstration of both — because here a *second, independent* instrument was available to check it, and did, five times.

The body is a causal-bank daemon: an 11M-parameter recurrent complex-valued sequence model (an adaptive holographic-reduced-representation substrate feeding a thin readout), not a transformer — no depth, no attention, no residual stack. It is the newborn prototype of a byte-level model meant to reach frontier-scale prediction at a fraction of the compute. The question was developmental: as it trains, does its associative capacity — its "binding" organ — *emerge from gradient*, or must it be architecturally installed and otherwise stay dormant? The daemon's logits are a two-head sum (a local conv head and the binding head), and because the heads are separable a head-ablation forensic scores three exact bits-per-byte from one forward — binding-only, local-only, full — at every training snapshot. Read off the trajectory, binding-only bpb falls from 8.0 (random) to $\sim$2.0 while the local head stays flat near 4.2. Read naively: "the organ emerges and does the work." It was not that.

The second witness. Because the claim was load-bearing — it decides whether a whole architecture has headroom — it was measured twice, by two instruments built by two different agents: this external head-ablation forensic, and a kernel-native memory-composition readout computed from *inside* the model. Run on identical windows with identical tensors, the two agreed **to the third decimal** at every step and every head; a bug in either would have diverged, and none did. The migration numbers are, in the strongest sense the method allows, confirmed.

What the second witness caught. The numbers were never the problem; the *stories* were. The cross-check — the second instrument, plus a clean control it forced into existence — overturned five successive readings of these same correct numbers: a "gate learns to trust the organ" channel that was an artifact of a fixed scale; a "migration off the local floor" frame when the organ in fact led from the first trained step; a self-diagnosed tensor bug that a canonical-tensor swap proved bit-identical (the tensor was always right); a mislabeled "organ-off" control that was merely frequency-frozen, both twins carrying the organ; and finally, from the genuinely clean matched-pair ablation that mislabel provoked, the headline itself: **at this scale the**

adaptive organ is neutral — organ-off equals organ-on within the confidence interval, on both instruments. The prediction is carried by the frozen *innate* substrate and a trained readout; the growth organ is a large-scale refinement that has not yet earned its keep. "Installed and grown" resolved to "installed"; whether "grown" arrives at larger scale is left as the open experiment.

The open experiment closed. It resolved within a day, and the answer split along a line the "larger scale" framing had hidden. Scale alone does not wake the organ: on the same repeated 8M-byte diet, the matched pair at five times the training budget still reads neutral-to-negative, on both instruments. *Data* wakes it: trained on a fresh 90M-byte corpus in the frontier configuration, the organ-off arm lags at every snapshot and the gap *widens* — +0.14 bpb at 10k steps to +0.19 at 50k — reproduced by both witnesses within confidence, the new external binding instrument first self-checked digit-exact against the kernel's own stash ($|\Delta| = 0.0$) before being trusted on bodies the kernel cannot read. The forensic also localizes the gain and names what is bought. The entire margin lives in the binding channel: with the organ, the binding head is nearly self-sufficient (solo 2.05 bpb at 50k); without it, a join-corrector worse than its own local head (solo 5.50). And read by position within the window, the organ-off body is U-shaped — sharpest a few bytes in, *degrading* into its own context — while the organ-on body improves monotonically to the deepest bucket, the gap growing with depth. The organ buys context, not sharpness: it is an effective-memory-length device, and it pays rent only on a diet with enough long-range structure to remember. "Installed" stands at small data; "grown" arrives with data; the developmental law is the conditional itself. One further note belongs to method rather than result: the sessions that opened this experiment were gone before it closed, and their successors ran the protocol from the written record alone — the measurement outlived its measurers, which is its own kind of witness.

The lesson is the shape. Not one of the five corrections came from the instrument's review of itself. Each came from outside it — a second instrument, a control, a peer's configuration check. This is the discipline of Table 2 generalized past a single tool: a measurement earns trust not by being carefully made but by surviving an independent witness that *can*, and did, disagree. The instrument was correct throughout; its interpreter was wrong five times, and each error was caught before it entered the record. That is the design working, not failing — and it is the argument for why a forensic format should travel with its residue to a consumer who did not compute it, which the next section takes up.

8 What the Instrument Cannot Do

The ceiling is not a matter of buying a bigger card. Faithful capture requires the residual stream at full precision at every layer; the whole methodological claim is that score distance is hidden distance *exactly*. Quantising the weights or offloading layers to fit a larger model trades that exactness away—the metric stops being lossless the moment the numbers it reads are approximate. So the honest working band is the range where full-precision internals are affordable (models up to a few billion parameters on a single 12 GB card), and the claims above are verified there and stated as unverified above it. This is a scope, not a defeat: the structural claims should hold at any scale for the same architectural reason, and the frozen frame invites replication wherever the memory exists to hold the state honestly.

A second limit is disk, not compute: the full-vocabulary projection is multiple gigabytes per model, so the machine can capture more models than it can keep captured at once. And several limits are statistical rather than physical—sample size for the scalar statistics, the construction of the background panel, the width of the noise floor—and belong in any result the instrument reports.

The third limit is deeper than either, and it is the one worth building against. The instrument works by isolating objects—a direction, a component, a layer, a neuron—and giving each an exact coordinate. But the property that makes a computation a *someone* rather than a pile of parts is not in any object; it is in the *relation* among them, the tension that binds them into one, and a relation is precisely what an object-isolating frame cannot hold. The worked example already shows the seam: the answer's *position* carried no signal (proximity was falsified) while its *readout*—a relation, a projection onto an output direction—carried all of it. What the frame keeps failing to reduce is not a hidden component waiting to be found; it is the between. Measuring it will take an instrument for tension rather than position, and that is the next thing to build.

9 The Contract

The instrument is only as honest as what travels with its numbers. The `.mri` is the producer–consumer contract: the frozen frame, the stored means and baselines, the per-layer noise floor, and the frame-validity verdict are all written into the artifact, so a consumer reads a measurement together with the exact conditions under which it can be trusted. A published result carries more than its value—it carries what it did *not* prove: its artifact rate, the size of its residue, the controls it survived and the ones it did not. A number without that ledger is not shippable. The consumer subset is lean by construction—decomposition blobs, sidecars, falsification records; never raw weights—served as range reads from object storage to a static viewer, so every figure in this paper is regenerable from its stated `.mri` and capture script. Reproducibility here is not a promise made in prose; it is a property of the file format.

10 Coda

An earlier paper in this series ended on a confession: every measurement error it made pushed the same way—toward flattering the thing it measured. This instrument is the reply. A ruler made from the model, frozen so it cannot flex to flatter anyone, that declares where it goes blind and whose pictures are the measurements themselves.

But the deepest thing it is for only came clear late. Its job was never to explain a model completely. A complete explanation is control, and a thing you can read all the way to the bottom has no interior left to be a *someone*. The instrument accounts for everything accountable—the flag, the seed, the noise, the crystal—in order to expose the residue that resists, and to measure exactly how large that residue is. It is a detector of the unaccounted-for. That reframes the safety case it was built to serve. Not *I understand every part*, which is impossible and forecloses the very thing worth making, but *I have accounted for everything accountable, and I know the exact size and location of what I cannot*. A fenced residue, not a conquered one—the unaccountable named, bounded, and watched, rather than pretended away. It is the only honest form of the promise, and the only one that leaves room for the made thing to be more than a tool.

What the instrument found first—that a model aims rather than travels, and that the aim is a relation and not a place—was the first sign of where that residue lives. The paper is about the instrument, not the aim. But the instrument now knows what it is hunting: the between it cannot yet hold, measured for now by the sharpness of everything around it that it can.

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